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Lab 03

Longitudinal Flight Dynamics

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0. Introduction

In this report, we will present the analysis and simulation of the longitudinal flight dynamics of a business jet.

This is extremely important in everyday applications, as it can help predict the behavior of the aircraft when it encounters turbulence or when it is doing climb and dive maneuvers.

In this laboratory, we will analyze and simulate the response of the aircraft to different inputs and we will compare the non-linear and linear response behaviors.

The study begins with the definition of the aircraft system and the aerodynamic parameters, then we will proceed with the solution of the trim problem to identify the steady-state conditions for straight-and-level flight.

After having verified the steady-state conditions, we will use MATLAB[®] and SIMULINK[®] to compute the non-linear dynamics of the system. In order to verify if it is necessary to use the more complex non-linear system, we will linearize the model and compare the results to define which method is more convenient in terms of computing costs and solution accuracy. The last part of the report will focus on the frequency-domain analysis and the linear system response to control surface perturbations.

This approach enables us to better understand the fundamental principles governing longitudinal aircraft dynamics and highlights the differences between non-linear and linear modeling techniques. The results serve as a basis for understanding the influence of aerodynamic parameters on flight stability and control.

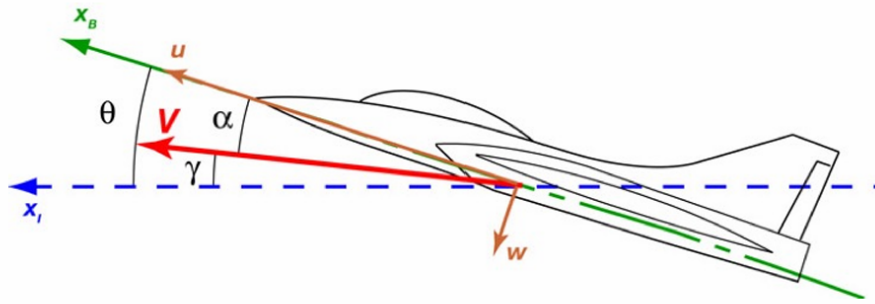


Figure 1: Aircraft in the body axes frame.

Definition of Variables

The following notation is used throughout and other variables not defined here will be introduced as needed in subsequent sections:

- T : Thrust acting along the body-axis x_B [N].
- u : Forward velocity component in the body frame [m/s].
- w : Vertical velocity component in the body frame [m/s].
- q : Pitch rate about the body-axis y [rad/s].
- θ : Pitch attitude angle [rad].
- h : Altitude [m].
- α : Angle of attack [rad].
- ρ : Air density [kg/m³].
- g : Gravitational acceleration [m/s²].
- δ_e : Elevator deflection [rad].

1. Task 1: Trim problem

Aircraft and Aerodynamic Model

In this task, the purpose is to numerically determine the trim condition — defined as a steady-state, straight-and-level flight — of an aircraft flying at sea level with an airspeed of $V_{\text{trim}} = 120$ knots. The aircraft model follows the standard longitudinal dynamics in the body-axis frame, and its motion is governed by the state vector (u, w, q, θ, h) .

The parameters characterizing the aircraft are:

- Mass: $m = 7484.4$ kg
- Pitch moment of inertia: $I_{yy} = 84309$ kg·m²
- Wing surface area: $S = 32.8$ m²
- Mean aerodynamic chord: $\bar{c} = 2.29$ m
- Thrustline angle: $\alpha_T = 0^\circ$
- Thrustline vertical distance from center of gravity: $\Delta z_T = 0.378$ m

At the trim altitude $h_{\text{trim}} = 0$ m (sea level), the atmospheric properties are obtained via the `atmosisa` function from the Aerospace Toolbox in MATLAB[®]. The dynamic pressure is computed as:

$$\bar{q} = \frac{1}{2} \rho V_{\text{trim}}^2$$

The aerodynamic coefficients are modeled as follows:

$$\begin{aligned} c_L &= c_{L_0} + c_{L_\alpha} \alpha + c_{L_{\delta_e}} \delta_e \\ c_D &= c_{D_0} + c_{D_\alpha} \alpha + c_{D_{\alpha^2}} \alpha^2 \\ c_M &= c_{M_0} + c_{M_\alpha} \alpha + c_{M_q} \left(\frac{\bar{c}}{V_{\text{trim}}} \right) q + c_{M_{\delta_e}} \delta_e \end{aligned}$$

The corresponding aerodynamic forces and moment are:

$$\begin{aligned} L(x) &= \bar{q} S c_L(x) \\ D(x) &= \bar{q} S c_D(x) \\ M(x) &= \bar{q} S \bar{c} c_M(x) \end{aligned}$$

Trim Condition Equations and Numerical Resolution

Regarding the trim equations, we started from the model of longitudinal dynamics in the body axes frame, which is described by the following system of equations:

$$\begin{cases} \dot{u} = -qw - g \sin \theta + \frac{1}{m} X_A + \frac{1}{m} X_T \\ \dot{w} = qu + g \cos \theta + \frac{1}{m} Z_A + \frac{1}{m} Z_T \\ \dot{q} = \frac{1}{I_{yy}} M_A + \frac{1}{I_{yy}} M_T \\ \dot{\theta} = q \\ \dot{h} = u \sin \theta - w \cos \theta \end{cases} \quad (1)$$

To find the trim condition, we assume a steady, level, non-accelerated flight. This leads to the following simplifications:

$$\dot{u} = \dot{w} = \dot{q} = \dot{h} = 0, \quad q = 0, \quad h = 0$$

We then rewrite the aerodynamic and thrust forces using the expressions below:

$$\begin{aligned}
X_A &= L \sin \alpha - D \cos \alpha \\
X_T &= T \cos \alpha_T = T \\
Z_A &= -L \cos \alpha - D \sin \alpha \\
Z_T &= -T \sin \alpha_T = 0 \\
M_A &= \frac{1}{2} \rho_0 V^2 S \bar{c} c_M \\
M_T &= \Delta z_T \cdot T = 0.38 \cdot T
\end{aligned}$$

These expressions are substituted into the simplified dynamics equations, resulting in the nonlinear system that defines the trim condition. The state vector is:

$$x = \begin{bmatrix} T_{\text{trim}} & \alpha_{\text{trim}} & \delta_{e_{\text{trim}}} & u_{\text{trim}} & w_{\text{trim}} & \theta_{\text{trim}} \end{bmatrix}^T$$

To fully constrain the system, we add two additional equations: the geometric relation between α , u , and w and the definition of the velocity magnitude

$$\alpha = \tan^{-1} \left(\frac{w}{u} \right), \quad V = \sqrt{u^2 + w^2}$$

The resulting nonlinear system is then written in vector form as:

$$F(x) = \begin{bmatrix} -g \sin \theta + \frac{1}{m} (L(x) \sin \alpha - D(x) \cos \alpha) + \frac{T}{m} \\ g \cos \theta + \frac{1}{m} (-L(x) \cos \alpha - D(x) \sin \alpha) \\ \frac{1}{I_{yy}} M(x) + \frac{1}{I_{yy}} \cdot 0.38 \cdot T \\ u \sin \theta - w \cos \theta \\ \alpha - \tan^{-1}(w/u) \\ V_{\text{trim}} - \sqrt{u^2 + w^2} \end{bmatrix} \quad (2)$$

Numerical Solution Procedure

To numerically solve the nonlinear system $F(x) = 0$ (2), we use the **fsolve** function from the Optimization Toolbox.

The aerodynamic coefficients c_L , c_D , and c_M are defined as functions of α and δ_e , and the lift, drag, and moment are computed accordingly. These are implemented as anonymous functions in MATLAB[®].

An initial guess $x_0 = \text{ones}(6, 1)$ is provided and passed to **fsolve**, which returns the solution x corresponding to the trim state:

$$(T_{\text{trim}}, \alpha_{\text{trim}}, \delta_{e_{\text{trim}}}, u_{\text{trim}}, w_{\text{trim}}, \theta_{\text{trim}})$$

This procedure gives the thrust, angle of attack, and elevator deflection needed to keep steady-level flight at the chosen speed and altitude.

2. Task 2: Trim verification (MATLAB®)

To verify that the trim state found in **Task 1** was actually a trim point for the airplane, we wrote a MATLAB® code to simulate the response of the aircraft, using the equations of longitudinal dynamics (1). After redefining the trim state vector as $x_{trim} = [u_{trim}; w_{trim}; q_{trim}; \theta_{trim}; h_{trim}]$, we set the initial conditions $x_0 = x_{trim}$ and the time interval at $[0 - 5]s$. The thrust and the elevator deflection are held at the calculated trim values. The MATLAB® code allowed us to plot the following diagrams:

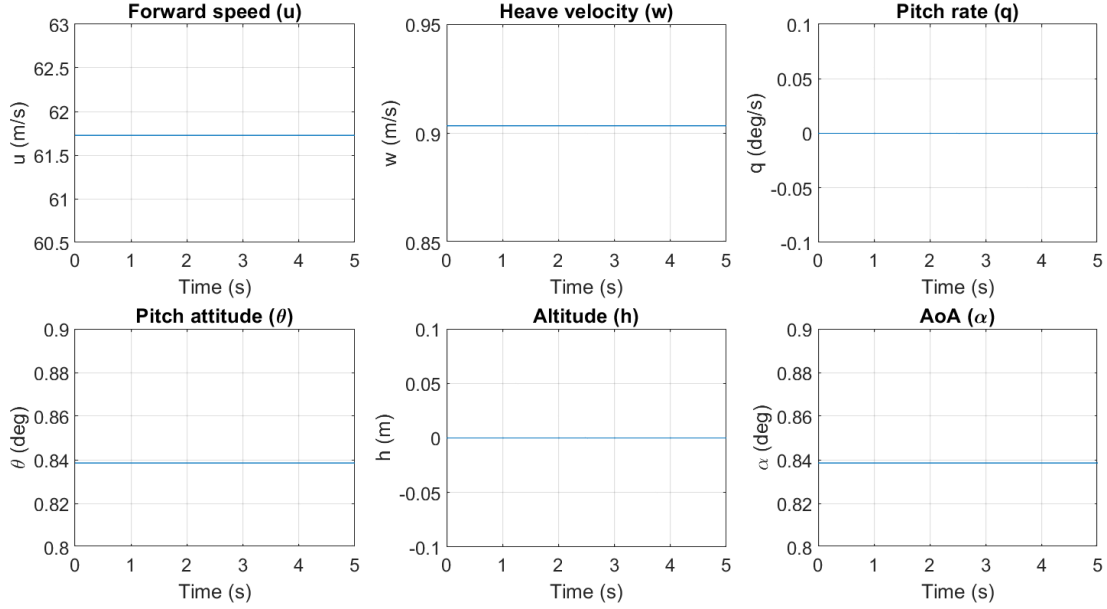


Figure 2: Trim verification diagrams.

As it can be observed in the image, since all the variables maintain a steady behavior, the trim is verified and is consistent with what we calculated in the previous Task. The trim solutions are:

$$x_{trim} = \begin{bmatrix} u_{trim} \\ w_{trim} \\ q_{trim} \\ \theta_{trim} \\ h_{trim} \end{bmatrix} = \begin{bmatrix} 61.7267 \\ 0.9034 \\ 0 \\ 0.0146 \\ 0 \end{bmatrix} \quad (3)$$

These values will be fundamental in the subsequent analysis of the airplane stability and are generally really useful for understanding the aircraft behavior as they offer a comparison to the stable equilibrium of the aircraft.

3. Task 3: Nonlinear response to elevator pulse (MATLAB[®])

In this task, we simulated the nonlinear longitudinal dynamics of the aircraft by integrating the equations presented in (1) with MATLAB[®] `ode45`, using the trim state vector $x_0 = x_{\text{trim}}$ (see (3) in Task 2) as the initial condition. The system was excited by a pulse input applied to the elevator deflection around the trim value (Figure 3):

$$\delta_e(t) = \delta_{e_{\text{trim}}} + \Delta\delta_e(t), \quad \Delta\delta_e(t) = \begin{cases} -1^\circ, & 5 \text{ s} < t < 10 \text{ s}, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

The aim of this analysis is to investigate how the aircraft responds to a temporary pitching moment perturbation caused by the deflection of the elevator. Figure 4 shows the time histories of the key state variables involved in longitudinal motion: forward velocity (u), vertical velocity (w), pitch rate (q), pitch angle (θ), altitude (h), and angle of attack (α).

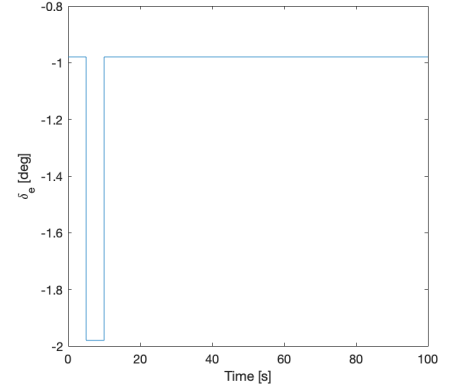


Figure 3: $\delta_e(t)$: Elevator Pulse Deflection

Comments on the Evolution of the Variables

Following the elevator pulse input, the aircraft exhibits a characteristic longitudinal dynamic response.

- **Forward velocity (u):** The forward speed decreases slightly immediately after the pulse and then oscillates with a long-period trend around the trim value. This is consistent with the relationship between pitch variations and airspeed in longitudinal dynamics.
- **Vertical velocity (w):** The vertical velocity increases sharply as the aircraft pitches up, reaching a peak shortly after $t = 5$ s, followed by damped oscillations. The fast transient indicates the immediate response of the aircraft to the control input.
- **Pitch rate (q):** The pitch rate shows a quick positive spike during the pulse input, reflecting the immediate nose-up rotation induced by the negative elevator deflection. A series of damped oscillations follows, indicating the natural short-period mode of the aircraft.
- **Pitch angle (θ):** The pitch angle increases immediately after the pulse, peaking around $t = 10$ s, and then exhibits slow, oscillatory behavior. The response clearly exhibits a combination of a fast short-period component and a slower, long-period variation in pitch.
- **Altitude (h):** The altitude shows a slow, sinusoidal variation, increasing in response to the nose-up attitude and then descending again.
- **Angle of attack (α):** The angle of attack reacts sharply to the elevator input, increasing quickly at the beginning of the pulse. After the initial peak, it settles into lightly damped oscillations around its nominal value, indicating the dynamic coupling between α and q .

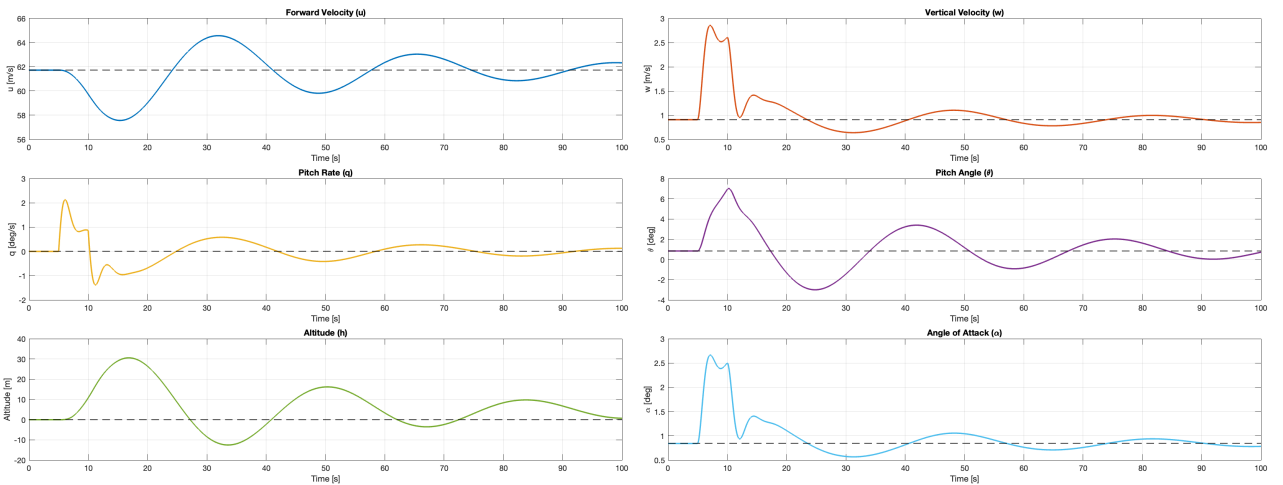


Figure 4: Nonlinear response to a -1° elevator pulse applied between $t = 5$ s and $t = 10$ s.

Overall, as shown in Figure 4, all state variables converge asymptotically to their trimmed equilibrium values, confirming the inherent stability of the longitudinal dynamics.

4. Task 4: Nonlinear response to elevator pulse (Simulink[®])

To obtain the non-linear response of the aircraft when the elevator is subjected to a one-degree pulse deflection from the trim value in the interval $[5 - 10]$ s we developed a Simulink[®] model based on **Task 3** and using an M-function.

The M-function takes as inputs the elevator step deflection u_{de} and the thrust at the trim condition T_{trim} , and computes the first derivative of the nonlinear state-space system using the equations in (1). This derivative is then integrated over time to update the state vector x . At each time step, the updated state vector x is fed back into the M-function to calculate the next state derivative. This feedback loop allows the system to simulate the nonlinear dynamics continuously.

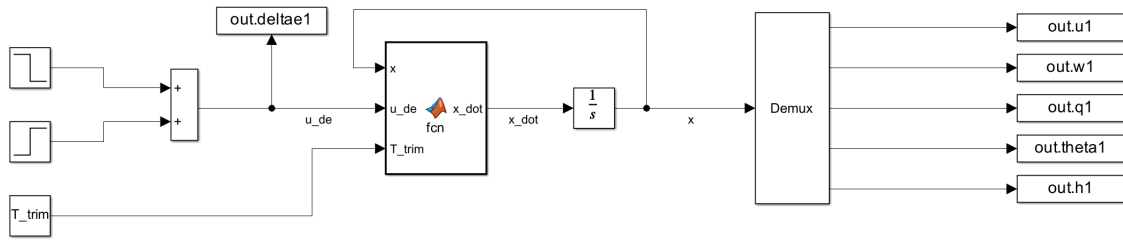
All meaningful parameters, including aircraft data and aerodynamic coefficients, are defined within the M-function.

Regarding the settings, the simulation was configured to start at $t = 0.0$ s and end at $t = 100.0$ s. The numerical integration was performed using the `ode45` solver, with a max step size of 0.01 to achieve smoother and more accurate simulation results and plots.

As for the model composition, the following blocks were utilized:

- **Step** blocks (used to create a one-degree pulse deflection u_{de} from 5 to 10 s)
- **Constant** block (T_{trim} used by the M-function)
- **Integral** block (configured with the trim conditions $x_0 = x_{trim}$, used to compute the system states over time by integrating their derivatives)
- **Demux** block (used to separate the outputs in order to send the values of u , w , q , θ and h to the workspace)
- **To Workspace** blocks (used to export the outputs to MATLAB[®])

The following image displays the Simulink[®] model described above:



Once the model was implemented, we generated the nonlinear response plots, which show the evolution of the state variables (u , w , q , θ and h) over time. In summary, the results confirm that the model accurately captures the essential characteristics of longitudinal dynamics, and the Simulink[®] outcome aligns with the theoretical expectations and the results obtained using the MATLAB[®] code alone.

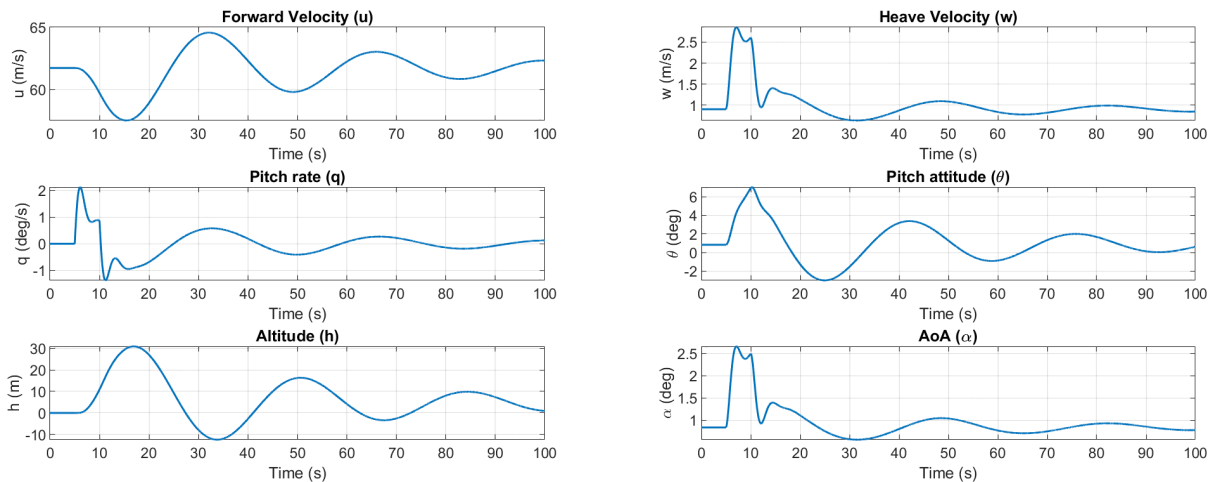


Figure 5: Nonlinear response to a -1° elevator pulse (Simulink[®] model).

5. Task 5: Linearized model – EOM

Now we derive the linearized longitudinal dynamics of the aircraft around the trim conditions previously found (3).

We will approximate the nonlinear equations of motion using a first-order Taylor expansion, resulting in a linear time-invariant (LTI) system suitable for control and stability analysis that can be expressed, as follows, in state-space form:

$$\dot{\mathbf{x}} = A_{\text{lon}}\mathbf{x} + B_{\text{lon}}\mathbf{u} \quad (5)$$

where the state vector $\mathbf{x}$ and input vector $\mathbf{u}$ are:

$$\mathbf{x} = \begin{bmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta \theta \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} \Delta \delta_e \\ \Delta T \end{bmatrix}$$

We start recalling the nonlinear model of longitudinal dynamics in the body axes frame:

$$\begin{cases} \dot{u} = -qw - g \sin \theta + \frac{1}{m}(X_A + X_T) \\ \dot{w} = qu + g \cos \theta + \frac{1}{m}(Z_A + Z_T) \\ \dot{q} = \frac{1}{I_{yy}}(M_A + M_T) \\ \dot{\theta} = q \\ \dot{h} = u \sin \theta - w \cos \theta \end{cases}$$

Each variable is decomposed into its equilibrium (trim) value corresponding to a steady-state straight-and-level flight and a small perturbation:

$$\begin{aligned} u &= u_0 + \Delta u, & w &= w_0 + \Delta w, & q &= q_0 + \Delta q, \\ \theta &= \theta_0 + \Delta \theta, & h &= h_0 + \Delta h \\ X &= X_0 + \Delta X, & Z &= Z_0 + \Delta Z, & M &= M_0 + \Delta M \end{aligned}$$

and we further make the distinction between the perturbation due to δ_e and to T :

$$\Delta X = \Delta X_A + \Delta X_T, \quad \Delta Z = \Delta Z_A + \Delta Z_T, \quad \Delta M = \Delta M_A + \Delta M_T$$

Aerodynamic forces and moments are linearized using a Taylor expansion of the first order around the trim point using aerodynamic stability and control derivatives, as indicated in pages 11 and 12 of the document [1]:

$$\begin{aligned} \Delta X_A &= X_u \Delta u + X_w \Delta w + X_q \Delta q + X_{\delta_e} \Delta \delta_e \\ \Delta Z_A &= Z_u \Delta u + Z_w \Delta w + Z_q \Delta q + Z_{\delta_e} \Delta \delta_e + Z_{\dot{w}} \dot{\Delta w} \\ \Delta M_A &= M_u \Delta u + M_w \Delta w + M_q \Delta q + M_{\delta_e} \Delta \delta_e + M_{\dot{w}} \dot{\Delta w} \\ \Delta X_T &= X_T \Delta T, \quad \Delta Z_T = Z_T \Delta T, \quad \Delta M_T = M_T \Delta T \end{aligned}$$

Linearized Equations of Motion

Substituting the perturbed variables and linearized forces into the nonlinear equations, we derive the linearized system.

- Longitudinal Velocity ($\Delta \dot{u}$):

$$\Delta \dot{u} = -(q_0 + \Delta q)(w_0 + \Delta w) - g \sin(\theta_0 + \Delta \theta) + X_u \Delta u + X_w \Delta w + X_{\delta_e} \Delta \delta_e + X_T \Delta T$$

- Vertical Velocity ($\Delta\dot{w}$):

$$\Delta\dot{w} = (q_0 + \Delta q)(u_0 + \Delta u) + g \cos(\theta_0 + \Delta\theta) + Z_u \Delta u + Z_w \Delta w + Z_q \Delta q + Z_{\dot{w}} \Delta\dot{w} + Z_{\delta_e} \Delta\delta_e + Z_T \Delta T$$

In this equation, the term involving $\Delta\dot{w}$ can be factored out on the left-hand side. To isolate $\Delta\dot{w}$ explicitly, we divide by $(1 - Z_{\dot{w}})$:

$$\Delta\dot{w} = \frac{1}{1 - Z_{\dot{w}}} [Z_u \Delta u + Z_w \Delta w + (Z_q + u_0) \Delta q + g \cos(\theta_0 + \Delta\theta) + Z_{\delta_e} \Delta\delta_e + Z_T \Delta T]$$

- Pitch Rate ($\Delta\dot{q}$):

$$\Delta\dot{q} = M_u \Delta u + M_w \Delta w + M_q \Delta q + M_{\dot{w}} \Delta\dot{w} + M_{\delta_e} \Delta\delta_e + M_T \Delta T$$

Since $\Delta\dot{w}$ appears in this equation, we substitute its expression obtained from the second equation:

$$\begin{aligned} \Delta\dot{q} &= M_u \Delta u + M_w \Delta w + M_q \Delta q + M_{\delta_e} \Delta\delta_e + M_T \Delta T \\ &\quad + M_{\dot{w}} \frac{1}{1 - Z_{\dot{w}}} [Z_u \Delta u + Z_w \Delta w + (Z_q + u_0) \Delta q + g \cos(\theta_0 + \Delta\theta) + Z_{\delta_e} \Delta\delta_e + Z_T \Delta T] \end{aligned}$$

- Pitch Angle ($\Delta\dot{\theta}$):

$$\Delta\dot{\theta} = q_0 + \Delta q$$

We neglect higher-order terms (such as products of perturbations such as $\Delta q \Delta w$) as they are assumed to be small. In addition, any quantities that are zero at the trim condition ($q_0 = 0$) are omitted to simplify the expressions.

By doing so, we can derive the linearized matrices. The state matrix A_{lon} and the input matrix B_{lon} are:

$$A_{\text{lon}} = \begin{bmatrix} X_u & X_w & -w_0 & -g \cos \theta_0 \\ \frac{Z_u}{1 - Z_{\dot{w}}} & \frac{Z_w}{1 - Z_{\dot{w}}} & \frac{Z_q + u_0}{1 - Z_{\dot{w}}} & \frac{-g \sin \theta_0}{1 - Z_{\dot{w}}} \\ M_u + \frac{M_{\dot{w}} Z_u}{1 - Z_{\dot{w}}} & M_w + \frac{M_{\dot{w}} Z_w}{1 - Z_{\dot{w}}} & M_q + \frac{M_{\dot{w}} (Z_q + u_0)}{1 - Z_{\dot{w}}} & \frac{-M_{\dot{w}} g \sin \theta_0}{1 - Z_{\dot{w}}} \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad B_{\text{lon}} = \begin{bmatrix} X_{\delta_e} & X_T \\ \frac{Z_{\delta_e}}{1 - Z_{\dot{w}}} & \frac{Z_T}{1 - Z_{\dot{w}}} \\ M_{\delta_e} + \frac{M_{\dot{w}} Z_{\delta_e}}{1 - Z_{\dot{w}}} & M_T + \frac{M_{\dot{w}} Z_T}{1 - Z_{\dot{w}}} \\ 0 & 0 \end{bmatrix} \quad (6)$$

We could reduce the matrix expressions even more by θ_0 little enough to approximate $\cos \theta_0 = 1$ and $\sin \theta_0 = 0$, and simplify the terms that include a parameter that is given as 0 in the table at page 14 of Lab03 – Longitudinal Flight Dynamics – part 2 by Lorenzo Dozio[1], which are X_q , $X_{\dot{w}}$, Z_q , $Z_{\dot{w}}$, $M_{\dot{w}}$ and X_{δ_e} .

6. Task 6: Analysis of the linear(ized) system

6.1. Natural frequency and damping ratio

We can evaluate the longitudinal dynamic characteristics by analyzing the stability of the aircraft longitudinal motion using the eigenvalues of the linearized state-space model evaluated at the trim condition. The state and input matrices A_{lon} and B_{lon} (6) can be derived in their numerical form using the numerical values of stability and control derivatives shown in the table at page 14 [1], and we find:

$$A_{\text{lon}} \approx \begin{bmatrix} -0.0571 & 0.1251 & -0.9034 & -9.8056 \\ -0.3051 & -0.8633 & 61.7267 & -0.1435 \\ -0.0015 & -0.0367 & -0.5442 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad B_{\text{lon}} \approx \begin{bmatrix} 0 & 1.3836 \times 10^4 \\ -7.3850 & 0 \\ -3.9097 & 5.2300 \times 10^3 \\ 0 & 0 \end{bmatrix} \quad (7)$$

A_{lon} gives us the following eigenvalues:

$$\lambda = \begin{cases} -0.7092 \pm 1.4989i \\ -0.0232 \pm 0.1863i \end{cases} \quad (8)$$

Since all eigenvalues have negative real parts, the equilibrium point is **asymptotically stable**. Each pair of complex conjugate eigenvalues represents a damped oscillatory mode of the system. If we write the generic complex eigenvalue as:

$$\lambda = \sigma \pm j\omega$$

The associated natural frequency ω_n and damping ratio ζ are then expressed as:

$$\begin{cases} \omega_n = \sqrt{\sigma^2 + \omega^2} \\ \zeta = -\frac{\sigma}{\omega_n} \end{cases} \quad (9)$$

By applying these to each pair of eigenvalues, we obtain the following pair of damped oscillatory modes:

- **Short period mode:**

Natural frequency: $\omega_{n1} = 1.6582$

Damping ratio: $\zeta_1 = 0.4277$

- **Phugoid mode:**

Natural frequency: $\omega_{n2} = 0.1877$

Damping ratio: $\zeta_2 = 0.1234$

The longitudinal motion consists of two distinct oscillations, a long-period oscillation called a phugoid mode and a short-period oscillation referred to as the short-period mode. The short period mode is characterized by a high-frequency, strongly damped oscillation, indicating a fast and well-damped response primarily associated with pitch rate and angle of attack dynamics. On the other hand, the phugoid mode is slower and lightly damped, representing an oscillation involving the exchange of kinetic and potential energy in longitudinal motion. Our values align well with the characteristics of the typical behavior of longitudinal aircraft dynamics.

6.2. Bode diagram

To analyze the dynamic influence of the elevator deflection δ_e on the longitudinal dynamics, we recall the linearized system in state-space form:

$$\dot{\mathbf{x}} = A_{\text{lon}}\mathbf{x} + B_{\text{lon}}\mathbf{u}$$

and we add the expression on the output:

$$\mathbf{y} = C_{\text{lon}}\mathbf{x} + D_{\text{lon}}\mathbf{u}$$

The output matrix C_{lon} and the feedthrough matrix D_{lon} are chosen to directly observe the variables of the longitudinal dynamics model in the frame of the body axes, with no feedthrough:

$$C_{\text{lon}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad D_{\text{lon}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (10)$$

From this representation, we compute the transfer function matrix $G(s)$ on MATLAB[®], which maps the input δ_e to the outputs u, w, q, θ .

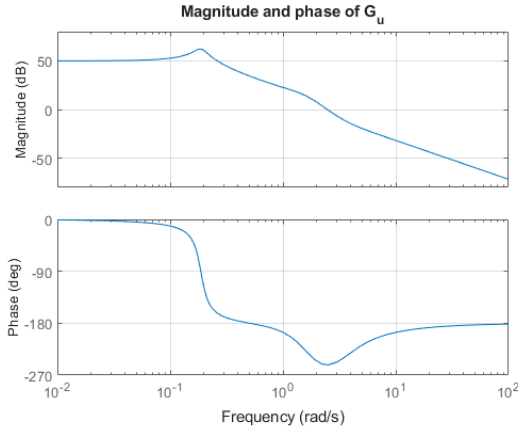
$$G_1(s) = \begin{bmatrix} G_{u\delta_e}(s) \\ G_{w\delta_e}(s) \\ G_{q\delta_e}(s) \\ G_{\theta\delta_e}(s) \end{bmatrix} \Rightarrow \begin{cases} G_{u\delta_e}(s) = \frac{\Delta u(s)}{\Delta \delta_e(s)} \\ G_{w\delta_e}(s) = \frac{\Delta w(s)}{\Delta \delta_e(s)} \\ G_{q\delta_e}(s) = \frac{\Delta q(s)}{\Delta \delta_e(s)} \\ G_{\theta\delta_e}(s) = \frac{\Delta \theta(s)}{\Delta \delta_e(s)} \end{cases}$$

where each transfer function is an element of the first column of the 4×2 transfer function matrix, given by:

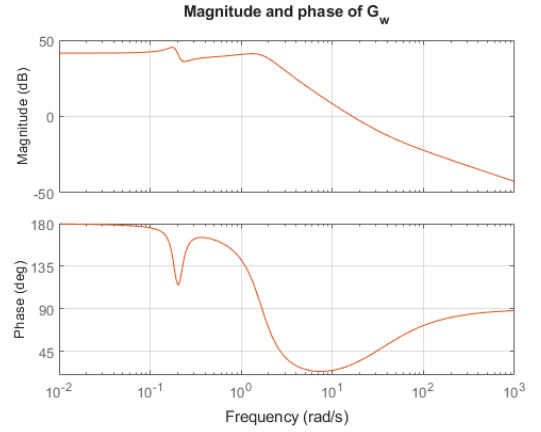
$$G(s) = C_{\text{lon}}(sI - A_{\text{lon}})^{-1}B_{\text{lon}} + D_{\text{lon}} \quad (11)$$

where s is the complex frequency variable for the Laplace transform, representing the behavior of the system in the frequency domain.

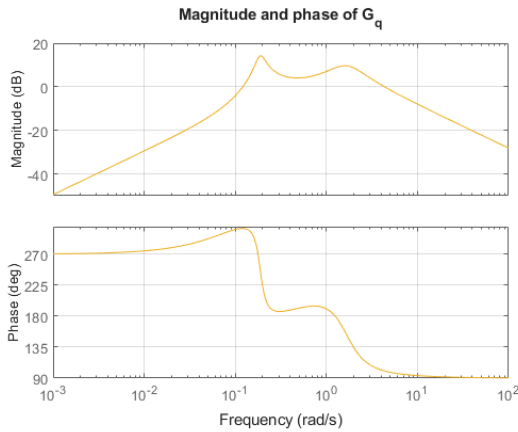
We can plot the magnitude and phase of the Bode diagrams of each transfer function to evaluate their behavior.



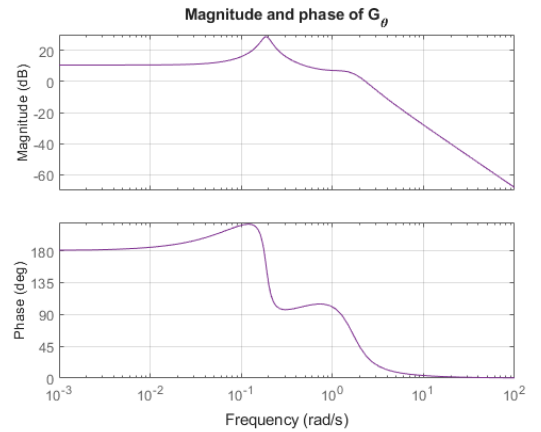
(a) Bode diagram of $G_{\delta_e, u}$



(b) Bode diagram of $G_{\delta_e, w}$



(c) Bode diagram of $G_{\delta_e, q}$



(d) Bode diagram of $G_{\delta_e, \theta}$

Figure 6: Bode diagrams of the transfer functions from elevator deflection δ_e to each output state.

The corresponding Bode plots of these transfer functions illustrate the frequency response of each state variable to the elevator input. In particular, the plots reveal the resonant peaks and phase lags associated with the short period and phugoid dynamics discussed in [Task 6.1](#). We will now analyze the effect of the phugoid and the short period mode on all the different state variables:

- **Forward velocity (u)**

The response of the forward velocity clearly shows the influence of the phugoid mode on it. The phugoid affects the long period oscillations in forward speed and altitude, in the higher frequencies the magnitude drops significantly which means that the forward speed does not respond to fast changes in elevator deflection. This reflects the inertia of the aircraft and its capability to adjust the forward speed in a short time-span.

- **Heave velocity (w)**

The heave velocity has a wider range of sensitivity. There is a small resonant peak which is due to the phugoid mode while a moderate influence from the short-period mode is also present. The elevator impulse can affect the heave velocity only indirectly, changing the pitch altitude and lift, which gets translated into vertical acceleration.

- **Pitch rate (q)**

The TF exhibits a strong resonance in the area of the natural frequency of the short-period mode. This is due to the fact that the elevator deflection directly impact on the pitching moment, hence the pitch rate. The sharp peak followed by a rapid attenuation at higher frequencies confirms that pitch rate dynamics are dominated by the short-period mode and the system does not sustain rapid pitch oscillations.

- **Pitch angle (θ)**

Since the pitch angle is an integrated measure of the pitch rate, it accumulates the effects of both modes. The bode plot for θ shows a broader frequency response, having contributions from the both the phugoid and the short-period dynamics. The magnitude is also relatively high at low frequencies, which shows that even a slow and small change in the elevator can lead to significant changes in pitch attitude. In the phase plot, instead, we can see the lag inherent to an integrated quantity, with an increased phase delay as the frequency rises.

We also display a combined Bode diagram of all four transfer functions with respect to δ_e for an at-a-glance comparison between the output behaviors.

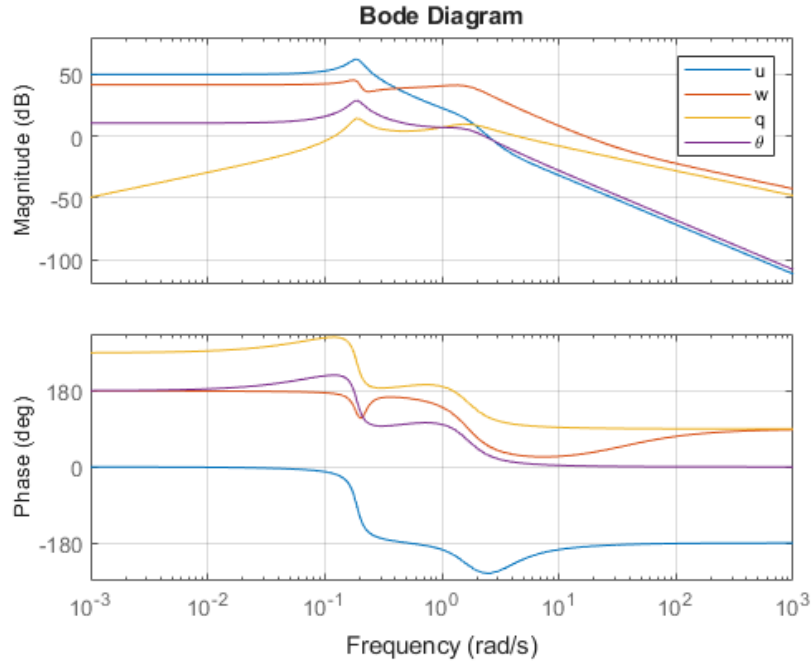


Figure 7: Bode diagram showing the frequency responses of u , w , q , and θ with respect to elevator deflection δ_e .

These responses inside the frequency domain can provide valuable insight into aircraft controllability and dynamic stability. The fact that we have distinct resonance peaks indicates that there are well-separated dynamic modes, which is a desirable

characteristic from a control design perspective. In this way, it is possible to significantly tune only on the short-period dynamic without affecting the phugoid too much.

Furthermore, the phase lag at higher frequencies highlights the system limitation in tracking fast control inputs, something to keep in mind when doing the control project for an autopilot or flight augmentation system. The system naturally filters out rapid oscillations, contributing to its robustness against high-frequency disturbances.

In conclusion, the bode diagrams shown above not only confirm the theoretical results we got from the eigenvalues analysis, but offer us additional information about the dynamic response of the aircraft to elevator inputs across the whole frequency spectrum.

7. Task 7: Linear response to elevator pulse (MATLAB®)

As in **Task 3**, the linearized longitudinal model was simulated under a -1° elevator pulse ($t = 5\text{--}10\text{ s}$, see (4) and **Figure 3**) by integrating the state-space system

$$\Delta\dot{x} = A_{\text{lon}} \Delta x + B_{\text{lon}} u(t),$$

with matrices A_{lon} and B_{lon} given in (6), using MATLAB's `ode45`.

The trim state vector $x_0 = x_{\text{trim}}$ from (3) served as the initial condition, and thrust perturbation was set to zero ($\Delta T(t) \equiv 0$).

The fifth state, altitude perturbation Δh , follows the kinematic relation

$$\Delta\dot{h} = u_0 \Delta\theta - \Delta w, \quad (12)$$

and the angle of attack at each time step was reconstructed as

$$\alpha(t) = \arctan(w(t)/u(t)).$$

Comments on the Linearized Response

Figure 8 presents the time histories of forward speed (u), heave velocity (w), pitch rate (q), pitch attitude (θ), altitude (h) and angle of attack (α).

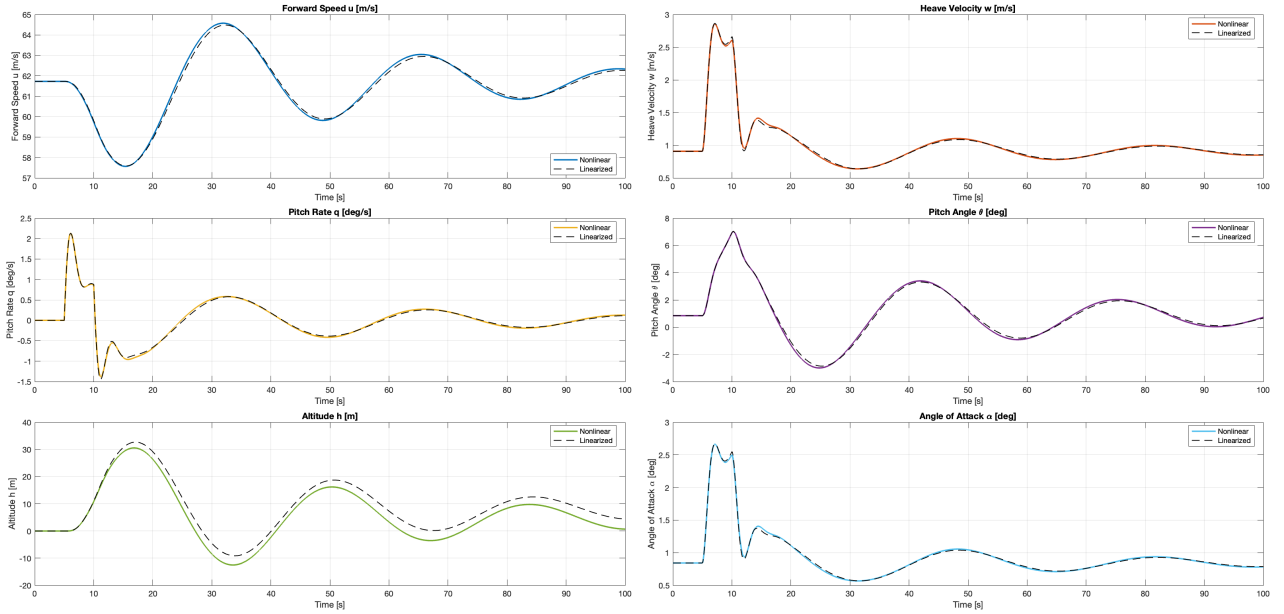


Figure 8: Linearized system response to a -1° elevator pulse.

The linearized model closely tracks the nonlinear response, accurately capturing both the fast short-period dynamics and the slower long-period trends.

The short-period oscillations in pitch rate and angle of attack align almost exactly in phase and frequency, while the forward and vertical velocity responses match the timing and envelope closely.

The only notable discrepancy is in altitude h , where the linear model slightly over-predicts the peak climb.

Overall, the linearized representation provides an excellent prediction of the longitudinal behavior of the aircraft for a -1° elevator pulse around the trim condition.

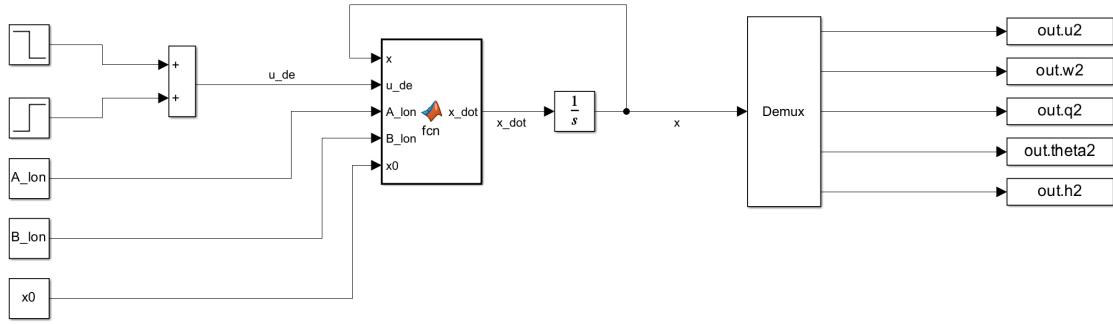
8. Task 8: Linear response to elevator pulse (Simulink®)

In order to analyze the linear response of the aircraft to a minus one-degree elevator pulse from its trim position within the interval $[5 - 10]$ s, a Simulink® model was developed based on the equations presented in [Task 7](#).

This model includes an M-function designed to compute the state derivatives of the linearized longitudinal motion. It takes as inputs the elevator deflection u_{de} , the linear state-space matrices A_{lon} and B_{lon} , and the trim condition x_0 . Based on the deviations from the trim state, the function returns the rate of change of the five system states: $\Delta u, \Delta w, \Delta q, \Delta \theta, \Delta h$, using the linearized equations of motion. Then, at each simulation step, the integrator numerically computes the state vector x by integrating its time derivative $\dot{x}$. The updated state is fed back into the function block, which evaluates a new $\dot{x}$ based on the current state. This iterative feedback process allows the system to evolve over time according to the linearized dynamics. Regarding the structure of the model, the following blocks were utilized:

- **Step** blocks (used to create a one-degree pulse deflection u_{de} from 5 to 10 s)
- **Three Constant** blocks (A_{lon} , B_{lon} and x_0 used by the M-function)
- **Integral** block (configured with the trim conditions x_0 , used to compute the system states over time by integrating their deviations $\Delta u, \Delta w, \Delta q, \Delta \theta, \Delta h$)
- **Demux** block (used to separate the outputs)
- **To Workspace** blocks (used to export the outputs to MATLAB®)

Below is the Simulink® model:



After the model was completed, the linear and non-linear responses were plotted and compared. The results confirm that the linearized model captures the main characteristics of the longitudinal dynamics with good accuracy. Moreover, the Simulink® simulation results are consistent with those obtained using the standalone MATLAB® code, validating the implementation and its alignment with theoretical expectations.

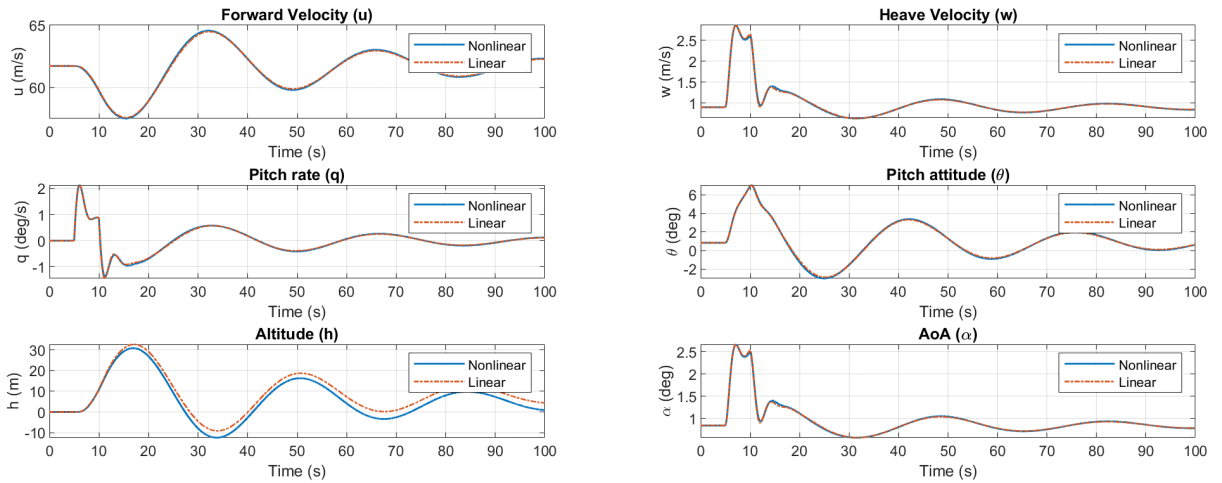


Figure 9: Linear and nonlinear response comparison (Simulink® model).

9. Conclusion

In this report, we were able to analyze and discuss the longitudinal flight dynamics of the aircraft. We observed that the undisturbed system is stable as the trim conditions are held over a relatively substantial period of time. Then we simulated the aircraft response to an elevator impulse using both MATLAB[®] and Simulink[®]. In order to have a comparison, we linearized the EOMs of the system and plotted the response, using this method, we realized that the linear version of the equations with small oscillations and limited time span is a good alternative to the non-linear version.

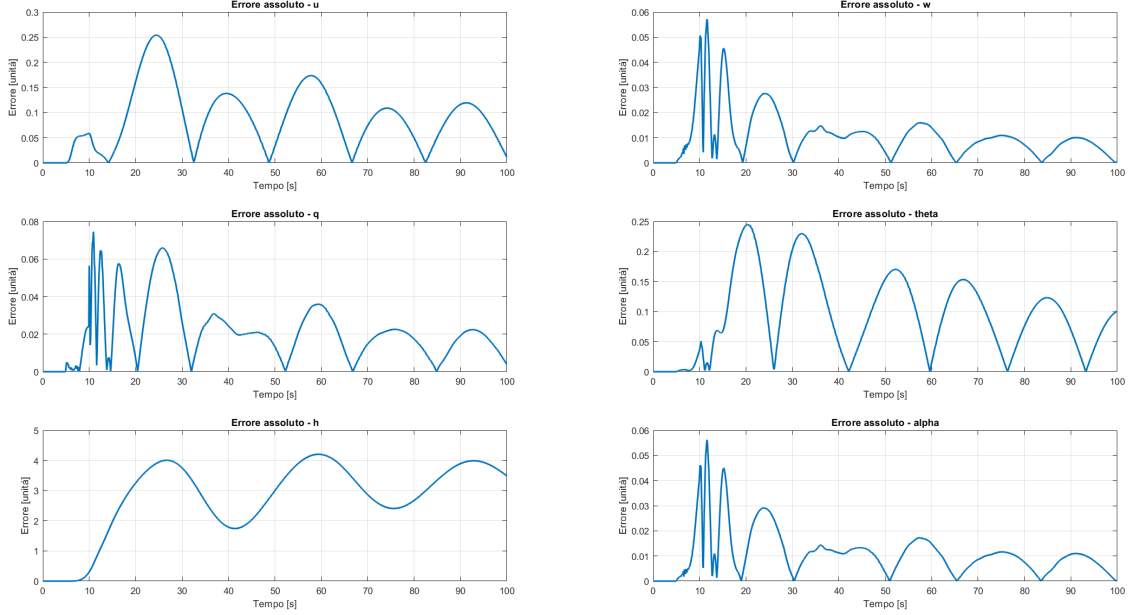


Figure 10: absolute error between linear and non-linear response

As we can observe from the image above, the absolute error between the linear and non-linear response is minimal. There is a general downward trend of the error for all variables except altitude.

The altitude, as seen in [Figure 8](#), is the variable that presents the biggest variation between the linear and non-linear model. There are two main purposes for which we use the altitude: to locate the airplane in the sky and to find the density of the air via the International Standard Air (ISA). Hence, the error is still under control since the variation is expressed in meters which does not imply a significant variation in terms of flight levels (100ft), in addition the density of the air does not vary much in such little intervals.

References

- [1] L. Dozio. *Lab03 – Longitudinal Flight Dynamics – part 2*. Politecnico di Milano, 2025 (cit. on pp. [8–10](#)).